



School of Informatics & IT
TEMASEK POLYTECHNIC

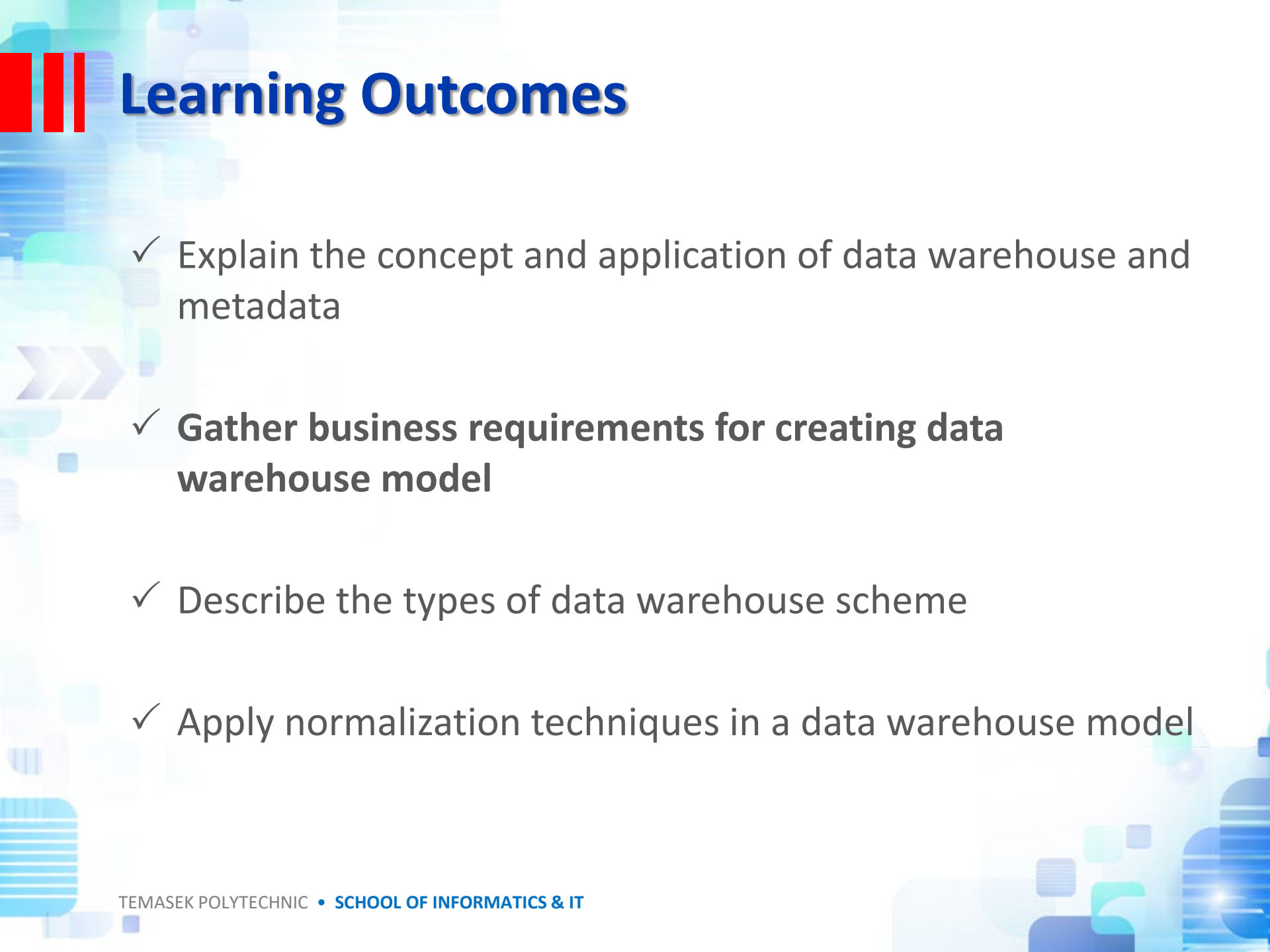
Data Warehousing and Business Intelligence

CDA2C01



Topic 2

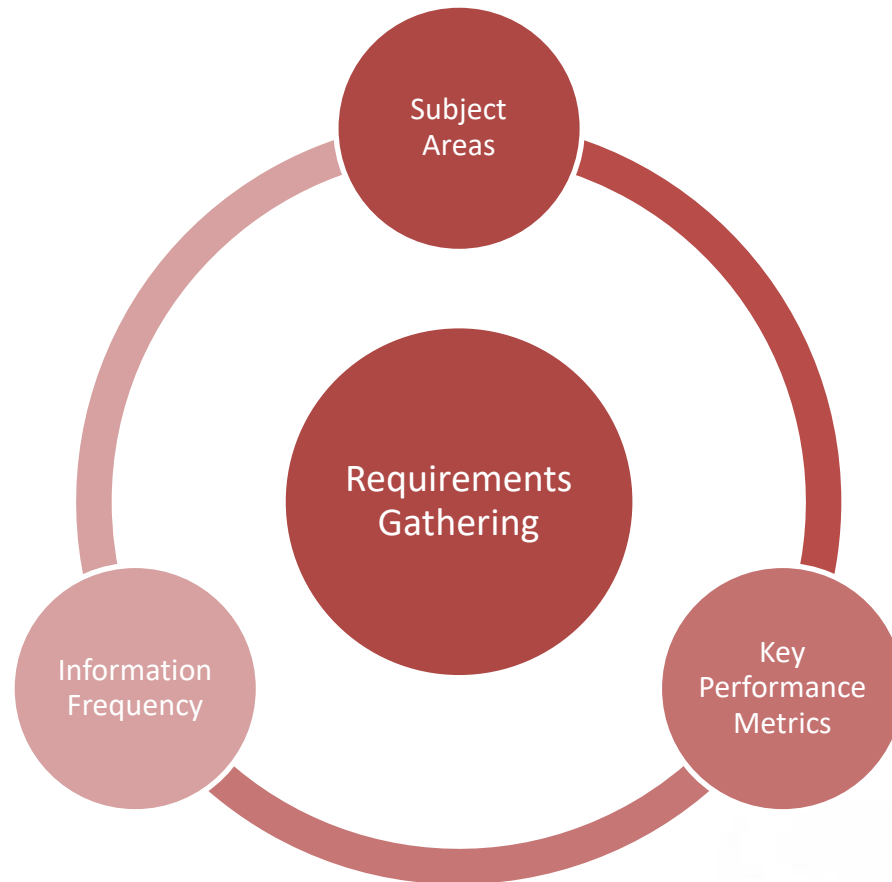
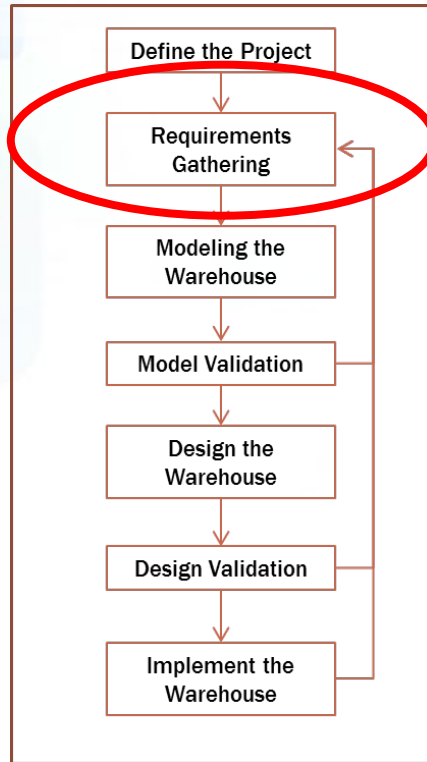
Introduction to Data Warehouse Model Design II



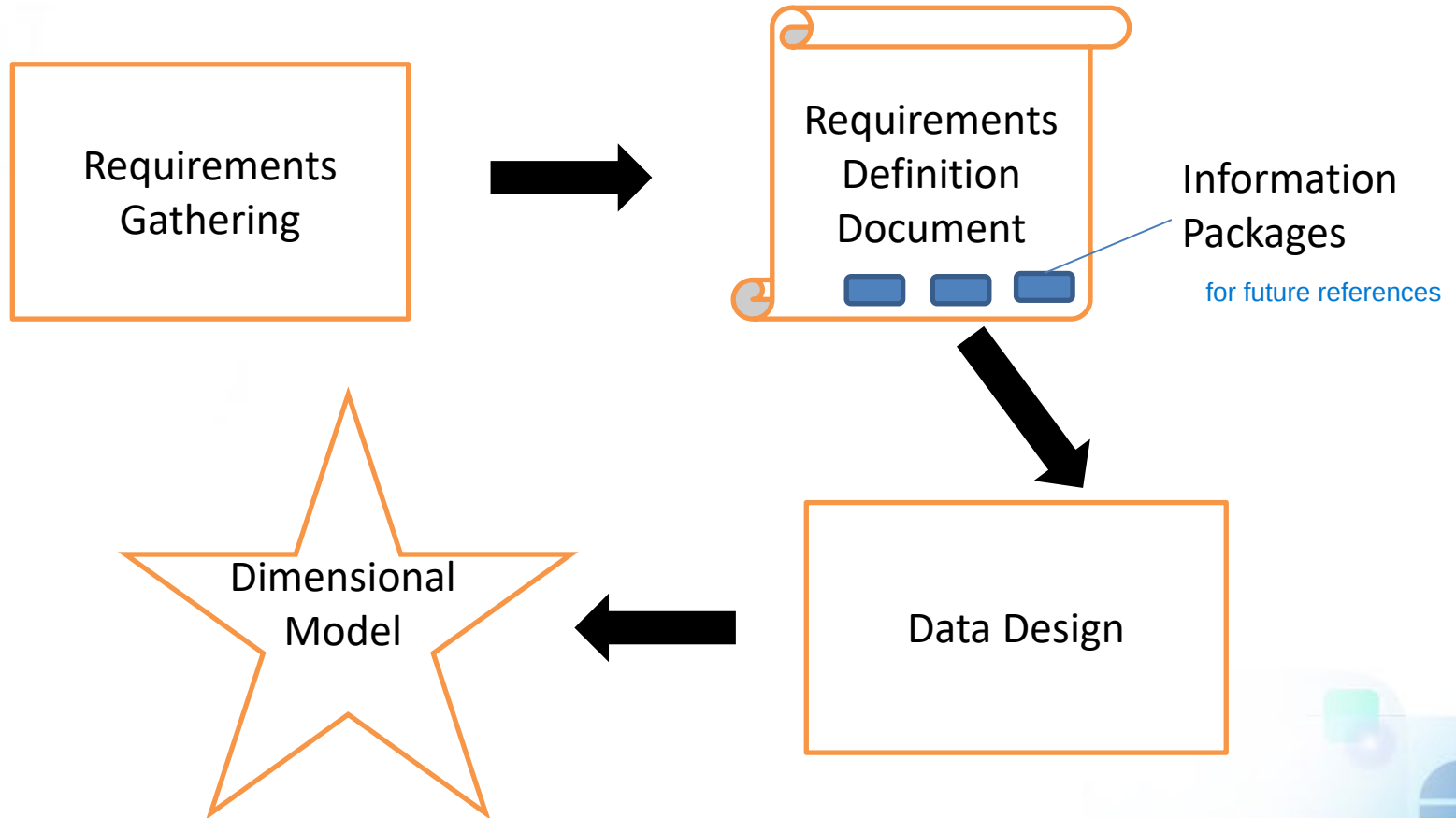
Learning Outcomes

- ✓ Explain the concept and application of data warehouse and metadata
- ✓ **Gather business requirements for creating data warehouse model**
- ✓ Describe the types of data warehouse scheme
- ✓ Apply normalization techniques in a data warehouse model

Considerations for Requirements Gathering



Requirements Gathering



Information Packages

Information packages are a technique to gather information requirements for a data warehouse. Information packages allow us to capture how the users think, in terms of metrics and how they are viewed along different business dimensions.

Essentially, information packages allow us to:

- Define the common subject areas
- Define the key business metrics
- Determine how data will be presented by their dimensions
- Determine how data will be aggregated or roll up
- Establish data granularity
- **Design the star schema**
- Estimate the data warehouse storage size
- Determine the frequency for data refreshing

Information Packages (example)

Information Subject: Sales Analysis

Dimensions

Hierarchies / Categories

Time	Location	Product	Customer
Year	Region	Brand	Age Group
Quarter	Country	Product Line	Income Group
Month	State	SKU	Gender
Day of week	Postal Code		Race
Holiday Flag	Street		
	City		
Facts: Forecast Sales, Budget Sales, Actual Sales			

Information Packages (example)

Information Subject: Car Sales

Dimensions

Hierarchies / Categories

Time	Product	Payment Method	Dealer	Customer
Year	Model Name	Finance Type	Dealer Name	Age Group
Quarter	Model Year	Term (months)	City	Income Group
Month	Product Line	Interest Rate	State	Gender
Day of week	Product Category	Agent Bank	Single Brand Flag	Race
Holiday Flag	Exterior Colour		Year First Operation	Own or Rent
	Interior Colour			Household Size
	First Year			Vehicles Owned
Facts: Actual Sale Price, Import Price, COE, Downpayment				

ER Modelling versus Dimensional Modelling

Entity-Relationship Modelling

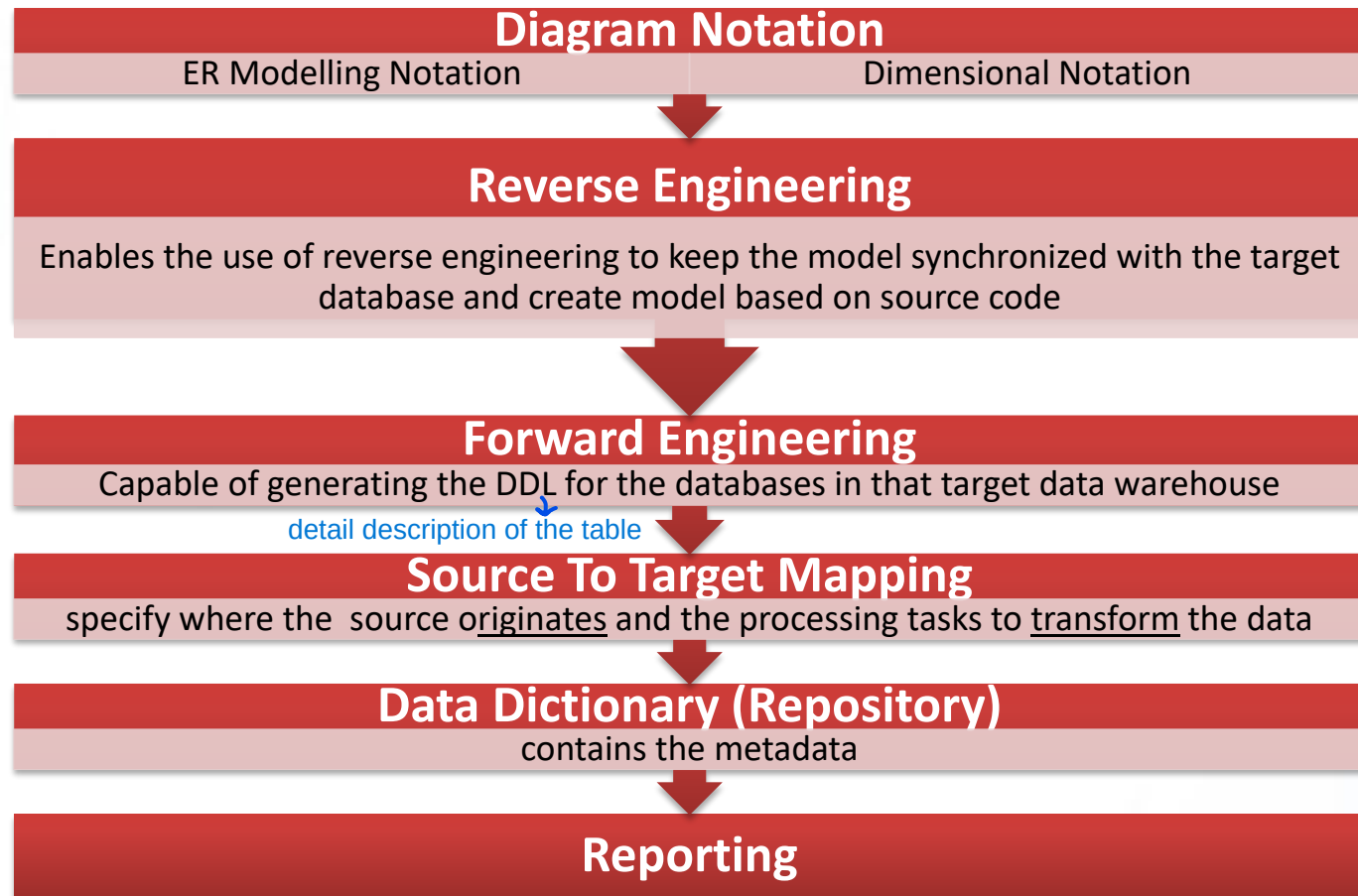
- OLTP systems capture details of events or transactions
- OLTP systems focus on individual events
- An OLTP system is a window into micro-level transaction
- Picture at detail level necessary to run the business
- Suitable only for questions at transaction level
- Data consistency, non-redundancy and efficient data storage critical

Dimensional Modelling

- DW meant to answer questions on overall process
- DW focus is on how managers view the business
- DW reveals business trends
- Information is centered around a business process
- Answers show how the business measures the process
- The measures to be studied in many ways along several business dimensions

Guidelines In Selecting A Data Modelling Tool

student — takes — subject } reverse engineering
taken



Data Modelling Tools

Examples of data modelling tools in the market

- 
- SQL Server Integration Services (SSIS) and SQL Server Analysis Services (SSAS)
 - MySQL Workbench
 - Oracle Designer
 - Sybase Power Designer

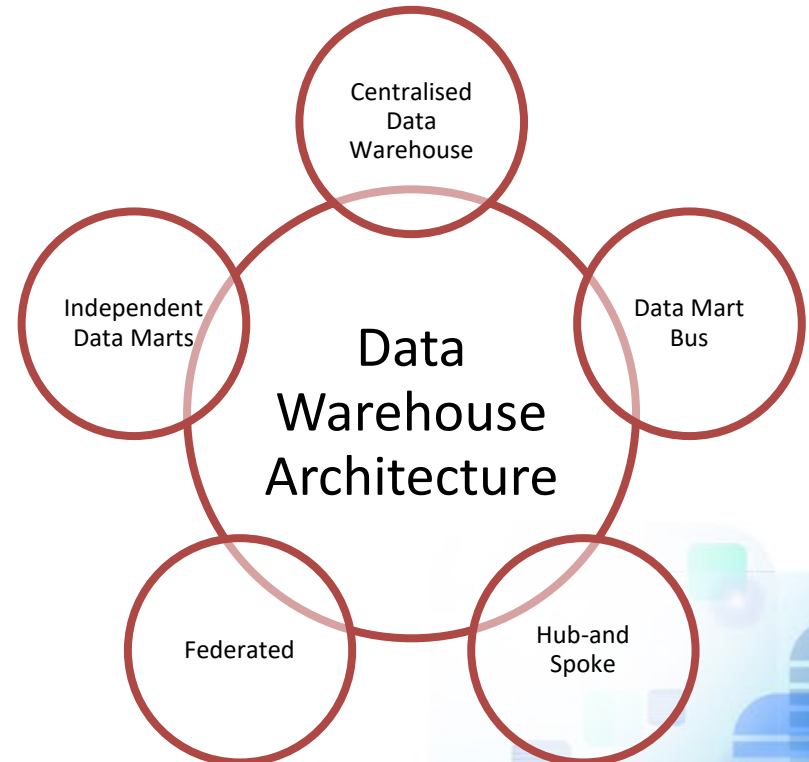


DATA WAREHOUSE ARCHITECTURE

Data Warehouse Architecture

Data warehouse architecture focuses on the data flows from source systems to end users.

There are five different architectural types according to Ariyachandra and Watson (2006):



Centralised Data Warehouse Architecture

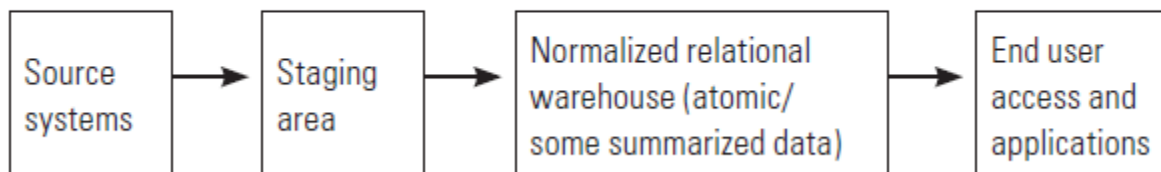
Takes into account the enterprise-level information requirements

An overall infrastructure is established from the onset

Data is usually stored in a normalised form at the lowest level of granularity

Queries and application access the data warehouse directly; there are no data marts

Centralized Data Warehouse Architecture

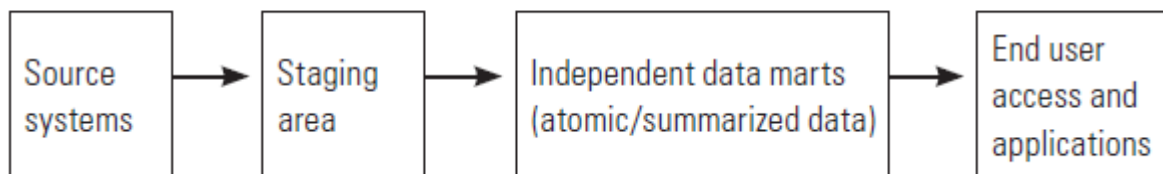


Independent Data Marts Architecture

The data marts are independent of one another – each data mart is created separately to serve the needs of a specific department/subject.

However, these separate data marts do not provide the organisation with a “single point of truth”. These data marts are likely to have inconsistent data definitions and standards. Such inconsistency hinders analysis of data across data marts.

Independent Data Marts Architecture



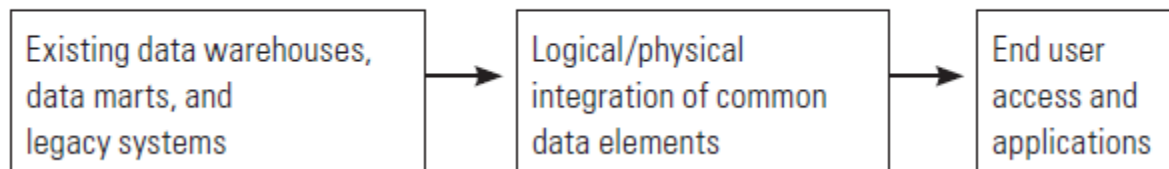
Federated Architecture

Some organisations have an existing legacy of decision-support systems, such as operational databases, extracted datasets and primitive data marts. It may not be prudent to discard all that huge investment and start from scratch.

Data is **loosely** integrated through shared key fields and globally defined metadata, with a heavy reliance on distributed queries. Data is fetched from disparate sources and then assembled at the user end-point.

In this architectural type, there is no one overall data warehouse

Federated Architecture



Hub-and-Spoke Architecture

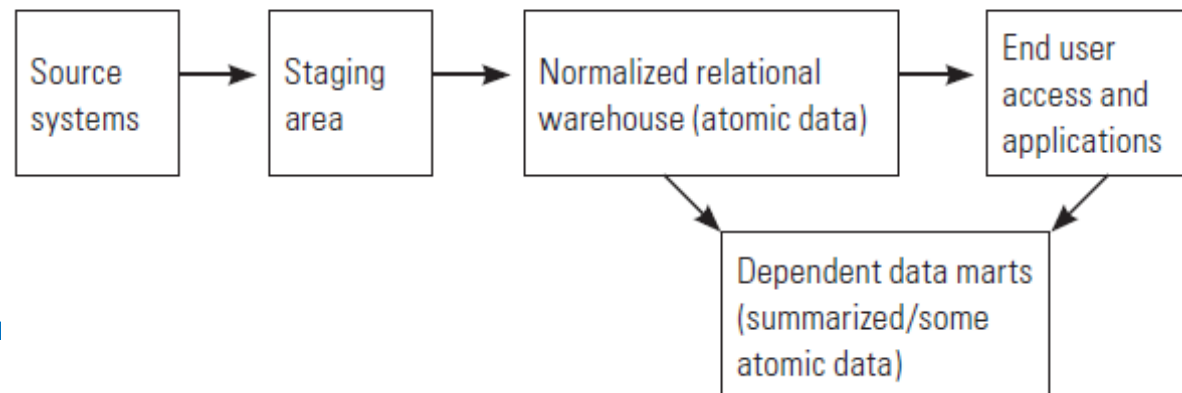
This is the **Corporate Information Factory (CIF)** approach by Bill Inmon. Similar to the centralised data warehouse architecture; data is stored in the centralised data warehouse in the normalised form at the lowest level of granularity.

The major difference is the presence of dependent data marts – the centralised data warehouse forms the hub to feed data to the data marts on the spokes.

Most queries are directed to the data marts, although the centralised data warehouse may itself be used for querying.

Hub-and-spoke results from a top-down approach to data warehouse development.

Hub and Spoke Architecture (Corporate Information Factory)



Data Mart Bus

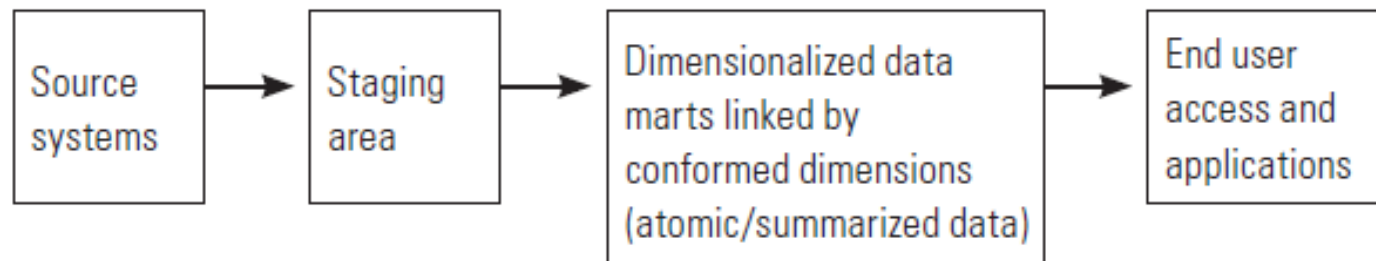
This is the **conformed dimensions** approach by Ralph Kimball

Start by creating data marts first. The key idea is that although the data marts are created separately, they are joined together by commonly defined dimensions (conforming to a set of business rules understood by the organisation). The conformed dimensions are also known as 'the bus'.

- The bus integrates the data marts to provide an enterprise view of the data

Data mart bus results from a bottom-up approach to data warehouse development

Data Mart Bus Architecture with Linked Dimensional Data Marts



Traditional Data Warehouse



2. Real-time data

1. Increasing data volumes

3. New data sources & type (non-relational data)

4. Cloud-born data

Modern Data Warehouse

BI and Analytics
(Self-service, Corporate, Collaboration, Mobile, Predictive)

Data Enrichment and Federated Query
(Single Query model, ETL, Data Quality, Master Data Management)

Data Management & Processing
(Relational, Non-Relational, Analytical, Streaming, Internal&External)

Data Source
(OLTP, ERP, CRM, LOB)

Non-relational Data
(Devices, Web, Sensors, Social)